Quantum Search Using Grover's Algorithm in Qiskit

Project Report

Name: Rahil Kothari
Course: Quantum Computing

Github Link - https://github.com/Rahilkothari/Quantum_Computing_Project

1. Abstract

This project presents the implementation of Grover's Quantum Search Algorithm using Qiskit. The objective is to manually construct the complete Grover circuit, including the oracle, diffusion operator, and iterative amplification process, without using built-in high-level functions.

The implementation uses a 10-qubit system representing 1024 possible states. Two target states are defined, and the algorithm amplifies their probability using multiple iterations. The simulation demonstrates how Grover's algorithm significantly increases the probability of finding the marked states compared to classical search.

2. Introduction

2.1 Background

Searching in an unstructured dataset is a fundamental problem in computer science. Classically, this requires O(N) time, meaning each element may need to be checked.

2.2 Grover's Algorithm

Grover's algorithm provides a quantum solution that reduces the complexity to $O(\sqrt{N})$, offering a quadratic speedup. It works using:

- Superposition of all states
- Oracle to mark target states
- Diffusion operator to amplify probabilities

2.3 Importance

Grover's algorithm is important in fields such as:

- Cryptography
- Optimization problems
- Quantum machine learning

It is a foundational concept in quantum computing.

3. Problem Statement

The project uses a 10-qubit system, giving a total of:

$N = 2^{10} = 1024$ states

Two marked states are defined:

- 0110011010
- 1101010001

The goal is to design a quantum circuit that increases the probability of measuring these states compared to others.

Constraint: The entire implementation must use basic Qiskit gates such as H, X, and MCX without using built-in Grover functions

4. Methodology

The implementation is divided into four main components:

4.1 Superposition

All qubits are initialized using Hadamard gates to create an equal probability distribution across all states.

Each state initially has equal probability.

4.2 Oracle

The oracle identifies target states by flipping their phase.

Steps:

1. Apply X gates based on target bits
2. Apply multi-controlled Z operation
3. Reverse X gates

This ensures only target states are affected.

4.3 Diffusion Operator

The diffusion operator amplifies the probability of marked states.

Steps:

1. Apply Hadamard gates
2. Apply X gates
3. Apply multi-controlled operation
4. Reverse X gates
5. Apply Hadamard again

This performs inversion about the mean.

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Diffusion gate count : 43
Diffusion depth      : 7

Diffusion circuit (text preview):
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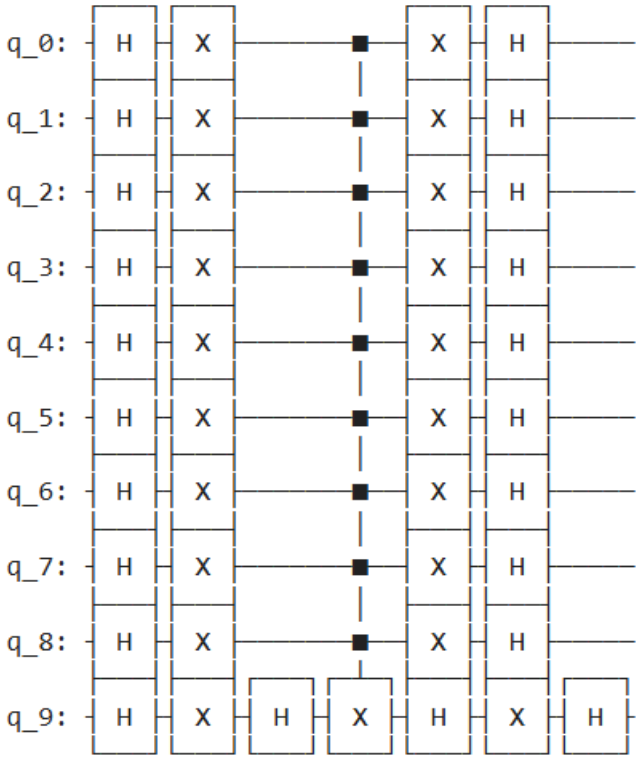

4.4 Grover Iterations

A Grover iteration consists of:

Oracle → Diffusion

The probability of finding target states increases with each iteration.

Optimal iterations are calculated using:

$k \approx (\pi/4) \times \sqrt{(N/M)}$

For this project:

- N = 1024
- M = 2
- Optimal k ≈ 17

5. Results and Observations

Simulations were performed using Qiskit AerSimulator with 4096 shots.

Observations:

- At 1 iteration: Small improvement in probability
- At 3 iterations: Noticeable amplification
- At 5 iterations: Strong amplification
- At 10 iterations: Significant probability (~64%)

The results closely match theoretical expectations.

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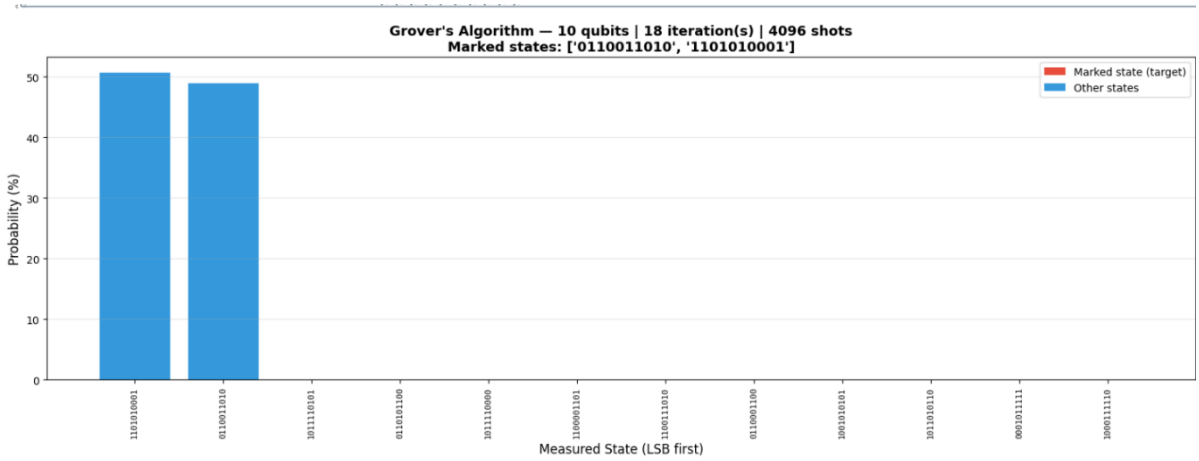
Simulation complete (4096 shots)
Unique outcomes measured: 12

── Marked state probabilities ──────────────────
  0110011010  (key: 0101100110)  →  0.0%  (0 / 4096 shots)
  1101010001  (key: 1000101011)  →  0.0%  (0 / 4096 shots)

── Top 10 measured states ──────────────────────
  1101010001  :  2079  ( 50.8%)
  0110011010  :  2007  ( 49.0%)
  1011110101  :     1  (  0.0%)
  0110101100  :     1  (  0.0%)
  1011110000  :     1  (  0.0%)
  1100001101  :     1  (  0.0%)
  1100111010  :     1  (  0.0%)
  0110001100  :     1  (  0.0%)
  1001010101  :     1  (  0.0%)
  1011010110  :     1  (  0.0%)
```

5.1 Visualization of Results



6. Analysis

6.1 Working Principle

Grover's algorithm works using:

- Quantum superposition
- Phase inversion
- Interference

These combine to amplify the probability of target states.

6.2 Amplitude Amplification

Each iteration rotates the system state toward the target states.

Probability follows:

$P = \sin^2((2k+1)\theta)$

6.3 Effect of Iterations

- Increasing iterations improves results initially
- Too many iterations reduce performance

6.4 Over-Rotation

If iterations exceed the optimal value, probability decreases again.
This shows the importance of choosing the correct number of iterations.

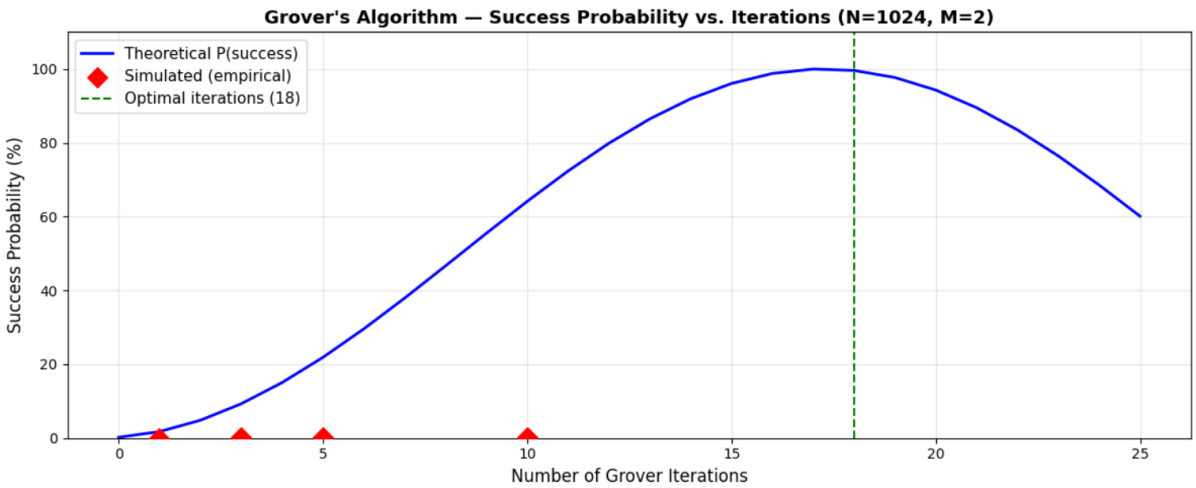
6.5 Quantum Advantage

Compared to classical search:

- Classical complexity: O(N)
- Quantum complexity: O(√N)

This provides a clear computational advantage.

6.6 Probability & Iteration Analysis

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Probability plot saved as 'grover_prob_vs_iterations.png'
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7. Conclusion

This project successfully demonstrates a manual implementation of Grover's Algorithm using Qiskit.

Key Achievements:

- Built full circuit using basic gates
- Implemented dynamic oracle
- Verified results through simulation
- Demonstrated quantum advantage

Key Learnings:

- Importance of amplitude amplification

- Understanding quantum interference
- Role of iteration count
- Practical challenges in implementation

Significance

Although current quantum hardware is limited, Grover's algorithm has strong future applications in optimization, cryptography, and search problems.

8. References

1. Grover, L. K. (1996). A fast quantum mechanical algorithm for database search.
2. IBM Quantum Learning Resources
3. Qiskit Documentation
4. Qiskit Aer Simulator Documentation